Student Portfolio Tutorial

Build & Publish Your Portfolio
Using VS Code and GitHub Pages

A Complete Step-by-Step Guide

From Setup to Deployment

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1 Tutorial Overview

Welcome to this comprehensive tutorial on creating and publishing your professional portfolio website! This guide will walk you through every step, from setting up your development environment to deploying your portfolio online.

Part	Topic	Time
Part 1	Setting Up Your Tools	20 min
Part 2	Creating Your Portfolio Project	30 min
Part 3	Building Your Portfolio	45 min
Part 4	Publishing to GitHub Pages	20 min
Part 5	Customization & Updates	15 min
	Total Estimated Time:	**~2 hours**

1.1 What You Will Learn

By the end of this tutorial, you will be able to:

- Set up VS Code with essential extensions for web development
- Create a professional, responsive portfolio website
- Use Git for version control
- Deploy your website using GitHub Pages for free
- Update and maintain your portfolio over time

1.2 Inspiring Portfolio Examples

Before you start building your own portfolio, take a look at some excellent examples created by Effat University engineering students! These portfolios will give you ideas and inspiration for your own design.

Effat University Engineering Student Portfolio Hub

Browse student portfolios for inspiration:

https://github.com/nailamarir/EffatUniversity-Engineering-Student-PortfolioHub

Tip

Explore the Portfolio Hub to see different styles, layouts, and approaches. Notice how each student showcases their unique skills and projects. Use these as inspiration, but make your portfolio reflect **your** personality and achievements!

1.3 Prerequisites

- A computer (Windows, Mac, or Linux)
- Internet connection
- Basic understanding of HTML/CSS (helpful but not required)

2 Part 1: Setting Up Your Tools

2.1 Step 1.1: Install Visual Studio Code

Visual Studio Code (VS Code) is a powerful, free code editor that we'll use throughout this tutorial.

Installation Steps

1. Go to https://code.visualstudio.com
2. Click **Download** for your operating system (Windows/Mac/Linux)
3. Run the installer and follow the installation wizard
4. Launch VS Code after installation

2.1.1 Recommended VS Code Extensions

After installing VS Code, add these helpful extensions to enhance your development experience:

1. Click the **Extensions** icon in the left sidebar (or press `Ctrl+Shift+X`)
2. Search and install each of the following:

Extension	Purpose
Live Server	Preview your website in real-time as you code
Prettier	Automatically format your code for consistency
HTML CSS Support	Better code completion for HTML and CSS
Auto Rename Tag	Automatically rename paired HTML tags

Tip

Live Server is especially useful! It automatically refreshes your browser whenever you save changes to your files.

2.2 Step 1.2: Install Git

Git is a version control system that tracks changes in your code and connects your local work to GitHub.

2.2.1 For Windows Users

1. Go to https://git-scm.com/download/win
2. Download and run the installer
3. Accept default settings (click Next through the wizard)
4. **Important:** Select "Git from the command line and also from 3rd-party software"

2.2.2 For Mac Users

Open Terminal and run:

```
xcode-select --install
```

2.2.3 Verify Installation

Open a terminal (or VS Code terminal with `Ctrl+‘`) and type:

```
git --version
```

You should see something like: `git version 2.40.0`

2.3 Step 1.3: Create a GitHub Account

GitHub is where you'll host your portfolio for free.

GitHub Account Setup

1. Go to https://github.com
2. Click **Sign Up**
3. Enter your email, create a password, and choose a username
4. Verify your email address

Important

Your username will be part of your portfolio URL!
For example, if your username is `saraahmed`, your portfolio will be at: `saraahmed.github.io`
Choose something professional (e.g., your name or a professional handle).

2.4 Step 1.4: Configure Git with Your Identity

Open VS Code Terminal (`Ctrl+‘`) and run these commands, replacing with your information:

```
git config --global user.name "Your Full Name"
git config --global user.email "your.email@example.com"
```

Verify your configuration:

```
git config --list
```

3 Part 2: Creating Your Portfolio Project

3.1 Step 2.1: Create Project Folder

3.1.1 Option A: Using File Explorer

1. Create a new folder on your computer named `my-portfolio`
2. Open VS Code
3. Go to **File** → **Open Folder** and select your `my-portfolio` folder

3.1.2 Option B: Using Terminal

```
mkdir my-portfolio
cd my-portfolio
code .
```

3.2 Step 2.2: Create Project Structure

In VS Code, create the following files and folders:

```
my-portfolio/
|
+-- index.html          <- Main HTML file
+-- css/
|   +-- style.css       <- Stylesheet
+-- js/
|   +-- script.js       <- JavaScript file
+-- images/             <- Folder for your images
    +-- profile.jpg     <- Your profile photo
```

3.2.1 How to Create Files/Folders in VS Code

1. **Right-click** in the Explorer panel (left sidebar)
2. Select **New File** or **New Folder**
3. Type the name and press Enter

   Or use the icons at the top of the Explorer panel:

   - New File
   - New Folder

4 Part 3: Building Your Portfolio

4.1 Step 3.1: Create the HTML Structure

Open `index.html` and add the following starter code:

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport"
          content="width=device-width, initial-scale=1.0">
    <title>Your Name - Portfolio</title>
    <link rel="stylesheet" href="css/style.css">
    <!-- Google Fonts -->
    <link href="https://fonts.googleapis.com/css2?
          family=Poppins:wght@300;400;500;600;700&display=swap"
          rel="stylesheet">
    <!-- Font Awesome Icons -->
    <link rel="stylesheet"
          href="https://cdnjs.cloudflare.com/ajax/libs/
          font-awesome/6.4.0/css/all.min.css">
</head>
<body>
    <!-- Navigation -->
    <nav class="navbar">
        <div class="nav-content">
            <a href="#" class="logo">Your Name</a>
            <ul class="nav-links">
                <li><a href="#home">Home</a></li>
                <li><a href="#about">About</a></li>
                <li><a href="#skills">Skills</a></li>
                <li><a href="#projects">Projects</a></li>
                <li><a href="#contact">Contact</a></li>
            </ul>
        </div>
    </nav>

    <!-- Continue with more sections... -->
</body>
</html>
```

4.1.1 Hero Section

Add this after the navigation:

```
<!-- Hero Section -->
<section id="home" class="hero">
    <div class="hero-content">
        <h1>Hi, I'm <span class="highlight">Your Name</span></h1>
        <p class="tagline">Computer Science Student |
           Aspiring Developer</p>
        <p class="intro">Welcome to my portfolio! I'm passionate
           about creating innovative solutions.</p>
        <div class="hero-buttons">
            <a href="#projects" class="btn btn-primary">
                View My Work</a>
            <a href="#contact" class="btn btn-secondary">
                Contact Me</a>
        </div>
    </div>
    <div class="hero-image">
        <img src="images/profile.jpg" alt="Your Name">
    </div>
</section>
```

4.1.2 About Section

```
<!-- About Section -->
<section id="about" class="about">
    <h2 class="section-title">About Me</h2>
    <div class="about-content">
        <div class="about-text">
            <p>Hello! I'm a Computer Science student at
               [Your University]. I'm currently in my [Year]
               year, focusing on [Your Interests].</p>
            <p>My journey in tech began when [Your Story].
               Since then, I've been constantly learning.</p>
        </div>
        <div class="about-info">
            <div class="info-item">
                <i class="fas fa-graduation-cap"></i>
                <span>BSc Computer Science</span>
            </div>
            <div class="info-item">
                <i class="fas fa-map-marker-alt"></i>
                <span>Jeddah, Saudi Arabia</span>
            </div>
        </div>
    </div>
</section>
```

4.1.3 Skills Section

```
<!-- Skills Section -->
<section id="skills" class="skills">
    <h2 class="section-title">My Skills</h2>
    <div class="skills-container">
        <div class="skill-category">
            <h3><i class="fas fa-code"></i>
                Programming Languages</h3>
            <div class="skill-tags">
                <span class="skill-tag">Python</span>
                <span class="skill-tag">Java</span>
                <span class="skill-tag">JavaScript</span>
                <span class="skill-tag">C++</span>
                <span class="skill-tag">SQL</span>
            </div>
        </div>
        <div class="skill-category">
            <h3><i class="fas fa-laptop-code"></i>
                Web Development</h3>
            <div class="skill-tags">
                <span class="skill-tag">HTML5</span>
                <span class="skill-tag">CSS3</span>
                <span class="skill-tag">React</span>
                <span class="skill-tag">Node.js</span>
            </div>
        </div>
    </div>
</section>
```

4.1.4 Projects Section

```
<!-- Projects Section -->
<section id="projects" class="projects">
    <h2 class="section-title">My Projects</h2>
    <div class="projects-grid">
        <div class="project-card">
            <div class="project-image">
                <img src="images/project1.jpg" alt="Project 1">
            </div>
            <div class="project-info">
                <h3>Project Title 1</h3>
                <p>A brief description of your project.</p>
                <div class="project-tech">
                    <span>Python</span>
                    <span>Flask</span>
                </div>
                <div class="project-links">
                    <a href="#"><i class="fab fa-github"></i>
                       Code</a>
                    <a href="#"><i class="fas
                       fa-external-link-alt">
                    </i> Demo</a>
                </div>
            </div>
```

```
            </div>
            <!-- Add more project cards as needed -->
        </div>
    </section>
```

4.1.5 Contact Section and Footer

```html
<!-- Contact Section -->
<section id="contact" class="contact">
    <h2 class="section-title">Get In Touch</h2>
    <div class="contact-content">
        <p>I'm currently looking for opportunities!</p>
        <div class="contact-methods">
            <a href="mailto:your.email@example.com"
               class="contact-item">
                <i class="fas fa-envelope"></i>
                <span>your.email@example.com</span>
            </a>
            <a href="https://linkedin.com/in/yourprofile"
               class="contact-item">
                <i class="fab fa-linkedin"></i>
                <span>LinkedIn</span>
            </a>
            <a href="https://github.com/yourusername"
               class="contact-item">
                <i class="fab fa-github"></i>
                <span>GitHub</span>
            </a>
        </div>
    </div>
</section>

<!-- Footer -->
<footer class="footer">
    <p>&copy; 2026 Your Name. All rights reserved.</p>
    <div class="social-links">
        <a href="#"><i class="fab fa-github"></i></a>
        <a href="#"><i class="fab fa-linkedin"></i></a>
        <a href="#"><i class="fab fa-twitter"></i></a>
    </div>
</footer>

<script src="js/script.js"></script>
```

4.2 Step 3.2: Add CSS Styling

Open `css/style.css` and add the following stylesheet. This creates a modern, responsive design.

4.2.1 CSS Variables and Reset

```
/* CSS Variables for Easy Customization */
:root {
    --primary-color: #6366f1;
    --primary-dark: #4f46e5;
    --text-color: #1f2937;
    --text-light: #6b7280;
    --bg-color: #ffffff;
    --bg-light: #f3f4f6;
    --gradient: linear-gradient(135deg, #6366f1 0%, #8b5cf6 100%);
    --shadow: 0 4px 6px -1px rgba(0, 0, 0, 0.1);
    --transition: all 0.3s ease;
}

/* Reset Styles */
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

html {
    scroll-behavior: smooth;
}

body {
    font-family: 'Poppins', sans-serif;
    line-height: 1.6;
    color: var(--text-color);
}
```

Tip

CSS Variables (custom properties) make it easy to change colors throughout your entire site by editing just a few lines at the top!

4.2.2 Navigation Styles

```
/* Navigation */
.navbar {
    position: fixed;
    top: 0;
    left: 0;
    right: 0;
    background: rgba(255, 255, 255, 0.95);
    backdrop-filter: blur(10px);
    box-shadow: var(--shadow);
    z-index: 1000;
    padding: 1rem 0;
}

.nav-content {
    max-width: 1200px;
    margin: 0 auto;
    padding: 0 2rem;
    display: flex;
    justify-content: space-between;
    align-items: center;
}

.logo {
    font-size: 1.5rem;
    font-weight: 700;
    color: var(--primary-color);
    text-decoration: none;
}

.nav-links {
    display: flex;
    list-style: none;
    gap: 2rem;
}

.nav-links a {
    font-weight: 500;
    text-decoration: none;
    color: var(--text-color);
    transition: var(--transition);
}

.nav-links a:hover {
    color: var(--primary-color);
}
```

4.2.3 Hero Section Styles

```
/* Hero Section */
.hero {
    min-height: 100vh;
    display: flex;
    align-items: center;
    justify-content: space-between;
    padding: 120px 10% 80px;
    background: var(--bg-light);
}

.hero-content {
    flex: 1;
    max-width: 600px;
}

.hero-content h1 {
    font-size: 3rem;
    font-weight: 700;
    margin-bottom: 1rem;
}

.highlight {
    background: var(--gradient);
    -webkit-background-clip: text;
    -webkit-text-fill-color: transparent;
}

.hero-image img {
    width: 350px;
    height: 350px;
    object-fit: cover;
    border-radius: 50%;
    border: 5px solid white;
    box-shadow: var(--shadow);
}

/* Buttons */
.btn {
    padding: 0.875rem 2rem;
    border-radius: 50px;
    font-weight: 600;
    transition: var(--transition);
    text-decoration: none;
    display: inline-block;
}

.btn-primary {
    background: var(--gradient);
    color: white;
}

.btn-secondary {
    border: 2px solid var(--primary-color);
    color: var(--primary-color);
}
```

4.2.4 Section and Card Styles

```
/* Section Styles */
section {
    padding: 100px 10%;
}

.section-title {
    font-size: 2.5rem;
    text-align: center;
    margin-bottom: 3rem;
}

/* Skills */
.skills-container {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(300px, 1fr));
    gap: 2rem;
}

.skill-category {
    background: white;
    padding: 2rem;
    border-radius: 15px;
    box-shadow: var(--shadow);
}

.skill-tag {
    background: var(--bg-light);
    padding: 0.5rem 1rem;
    border-radius: 25px;
    display: inline-block;
    margin: 0.25rem;
}

/* Projects Grid */
.projects-grid {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(350px, 1fr));
    gap: 2rem;
}

.project-card {
    background: white;
    border-radius: 15px;
    overflow: hidden;
    box-shadow: var(--shadow);
    transition: var(--transition);
}

.project-card:hover {
    transform: translateY(-10px);
}
```

4.2.5 Responsive Design

```
/* Responsive Design for Mobile */
@media (max-width: 768px) {
    .nav-links {
        display: none;
    }

    .hero {
        flex-direction: column-reverse;
        text-align: center;
        padding-top: 100px;
    }

    .hero-content h1 {
        font-size: 2rem;
    }

    .hero-image img {
        width: 250px;
        height: 250px;
        margin-bottom: 2rem;
    }

    .about-content {
        grid-template-columns: 1fr;
    }

    .footer {
        flex-direction: column;
        gap: 1rem;
        text-align: center;
    }
}
```

Note

The `@media` query ensures your portfolio looks great on mobile devices. The styles inside only apply when the screen is 768 pixels wide or smaller.

4.3 Step 3.3: Add JavaScript

Open `js/script.js` and add the following code for interactivity:

```
// Smooth Scrolling for Navigation Links
document.querySelectorAll('a[href^="#"]').forEach(anchor => {
    anchor.addEventListener('click', function (e) {
        e.preventDefault();
        const target = document.querySelector(
            this.getAttribute('href'));
        if (target) {
            target.scrollIntoView({
                behavior: 'smooth',
                block: 'start'
            });
        }
    });
});

// Navbar Background Change on Scroll
const navbar = document.querySelector('.navbar');

window.addEventListener('scroll', () => {
    if (window.scrollY > 100) {
        navbar.style.background = 'rgba(255, 255, 255, 0.98)';
    } else {
        navbar.style.background = 'rgba(255, 255, 255, 0.95)';
    }
});

console.log('Portfolio loaded successfully!');
```

4.4 Step 3.4: Preview Your Website

Using Live Server

1. Right-click on `index.html` in VS Code
2. Select **"Open with Live Server"**
3. Your browser will open with your portfolio!
4. Changes you make will automatically refresh

5 Part 4: Publishing to GitHub Pages

5.1 Step 4.1: Create a GitHub Repository

Create Your Repository

1. Go to https://github.com and log in
2. Click the + icon in the top right → **New repository**
3. **Repository name:** `yourusername.github.io`
4. Set to **Public**
5. Leave other options unchecked
6. Click **Create repository**

Important

Replace `yourusername` with your actual GitHub username. This special naming convention (`username.github.io`) enables GitHub Pages automatically!

5.2 Step 4.2: Initialize Git in Your Project

Open VS Code Terminal (`Ctrl+‘`) in your project folder:

```
# Initialize a new Git repository
git init

# Add all files to staging
git add .

# Create your first commit
git commit -m "Initial commit: My portfolio website"
```

5.3 Step 4.3: Connect to GitHub and Push

Replace `yourusername` with your GitHub username:

```
# Connect to your GitHub repository
git remote add origin
    https://github.com/yourusername/yourusername.github.io.git

# Push your code to GitHub
git branch -M main
git push -u origin main
```

5.3.1 Authentication Options

If prompted for credentials, you have two options:

Option A: Personal Access Token (Recommended)

1. Go to GitHub → Settings → Developer Settings
2. Click Personal Access Tokens → Tokens (classic)
3. Generate new token with `repo` scope
4. Use this token as your password when pushing

Option B: GitHub CLI

```
gh auth login
```

5.4 Step 4.4: Enable GitHub Pages

Enable GitHub Pages

1. Go to your repository on GitHub
2. Click **Settings** (tab at the top)
3. In the left sidebar, click **Pages**
4. Under **Source**, select:
   - Branch: `main`
   - Folder: `/ (root)`
5. Click **Save**
6. Wait 1-2 minutes for deployment

5.5 Step 4.5: Access Your Live Portfolio!

Congratulations!

Your portfolio is now live at:

`https://yourusername.github.io`

6 Part 5: Customization & Updates

6.1 Making Changes to Your Portfolio

Whenever you want to update your portfolio:

```
# 1. Make your changes in VS Code

# 2. Stage your changes
git add .

# 3. Commit with a descriptive message
git commit -m "Updated projects section"

# 4. Push to GitHub
git push
```

Note

Changes will be live within 1-2 minutes after pushing!

6.2 Customization Ideas

6.2.1 Change Color Theme

Edit the CSS variables in `style.css`:

```
:root {
    --primary-color: #your-color-here;
    --secondary-color: #your-color-here;
}
```

6.2.2 Popular Color Palettes

Theme	Primary	Secondary
Ocean	#0ea5e9	#06b6d4
Forest	#22c55e	#10b981
Sunset	#f97316	#ef4444
Royal	#8b5cf6	#a855f7

6.2.3 Add Your Profile Photo

1. Add your photo to the `images` folder
2. Name it `profile.jpg`
3. Make sure it's square (same width and height)

7 Troubleshooting

Problem	Solution
"Permission denied" when pushing	Set up SSH keys or use a Personal Access Token
Changes not showing on live site	Wait 2-5 minutes; try hard refresh (Ctrl+Shift+R)
Images not loading	Check file paths are correct and case-sensitive
Site shows 404 error	Ensure repository name is exactly `yourusername.github.io`

8 Additional Resources

8.1 Effat Student Portfolio Examples

Effat University Engineering Student Portfolio Hub

https://github.com/nailamarir/EffatUniversity-Engineering-Student-PortfolioHub

Explore portfolios created by your fellow students for inspiration!

8.2 Useful Links

- **GitHub Pages Documentation:** https://docs.github.com/en/pages
- **VS Code Documentation:** https://code.visualstudio.com/docs
- **HTML Reference:** https://developer.mozilla.org/en-US/docs/Web/HTML
- **CSS Reference:** https://developer.mozilla.org/en-US/docs/Web/CSS
- **Font Awesome Icons:** https://fontawesome.com/icons
- **Google Fonts:** https://fonts.google.com

9 Completion Checklist

☐	Installed VS Code
☐	Installed Git
☐	Created GitHub account
☐	Configured Git identity
☐	Created project structure
☐	Added HTML content
☐	Added CSS styling
☐	Added JavaScript
☐	Tested locally with Live Server
☐	Created GitHub repository
☐	Pushed code to GitHub
☐	Enabled GitHub Pages
☐	Verified live site works

Congratulations!

You've successfully created and published your portfolio website!

Keep updating it with new projects, skills, and experiences as you grow in your career!

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